Day 3 - Memory Fundamentals Mission

Definition of Done

By the end of today, every team member should be able to answer:

1. Where does data live while a program runs?
2. What is RAM?
3. What is the Stack?
4. What is the Heap?
5. Why does memory usage increase when applications run?
6. Why does a computer become slow when many applications are open?

You are NOT allowed to memorize definitions.

Your goal is to SEE memory in action.

Part 1 - Create a Mental Movie

Imagine:

Storage (SSD) ↓ RAM ↓ CPU

Example:

Chrome exists inside SSD.

When Chrome is opened:

1. Operating System copies Chrome into RAM.
2. CPU executes instructions from RAM.
3. Variables are stored in memory.
4. Objects are created in memory.

Question:

If RAM is full, where will new data go?

Think before searching.

Write your answer.

Part 2 - Experiment 1

Create a new HTML file.

Paste:

let name = "Harsha";

console.log(name);

Open browser.

Open DevTools.

Observe:

- Console
- Sources

Question:

Where does the variable exist while the page is running?

Write your thoughts.

Part 3 - Experiment 2

Replace code with:

const users = [];

for(let i = 0; i < 1000; i++){ users.push({ id: i, name: "User " + i }); }

console.log(users);

Refresh page.

Observe:

Browser still works normally.

Question:

What happened in memory?

Write your prediction.

Part 4 - Experiment 3

Increase load.

Replace:

1000

with:

100000

Run again.

Observe:

- Browser responsiveness
- CPU usage
- Memory usage

Open:

Task Manager

Find browser process.

Record:

Memory Before

Memory After

Question:

Why did memory increase?

Write answer.

Part 5 - Learn the Concepts

Study only these topics:

1. RAM
2. Stack
3. Heap

Do NOT go into advanced memory management.

Only understand purpose.

Understanding RAM

Answer:

What happens when an application starts?

Understand:

- RAM stores currently running data.

- RAM is fast.
- RAM is temporary.
- Data disappears when power is removed.

Understanding Stack

Think:

Stack stores simple, short-lived values.

Example:

let age = 20;

Question:

Why can the computer find age quickly?

Write your own answer.

Understanding Heap

Think:

Heap stores larger, dynamic objects.

Example:

const user = { name: "Harsha", age: 20 };

Question:

Why can't large objects be handled exactly like simple values?

Write your own answer.

Part 6 - Research

Why does a computer become slow when many applications are open?